Effect of Household Insecticide Use on Health Outcomes

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1 Introduction

In the landmark book *Silent Spring*[1], Rachel Carson introduces the reader to the perils of dichlorodiphenyltrichloroethane, a synthetic organochlorine insecticide (DDT) used for pest control. DDT not only disrupts the balance of the ecosystems, but causes widespread death and negative health outcomes in animals and people alike. Although this exposé successfully warned the American consumer and launched the environmentalist movement, DDT is still used along with other household insecticides.

In a previous work entitled "The Association between organosphosphate insecticides, blood pressure dysregulation, and metabolic syndrome among U.S. Adults: NHANES 2015-2016"[2] it was found that widely used organophosphate insecticides were associated with negative effects of human health including a decrease in mean systolic and diastolic blood pressure. Although the findings of this study are interesting, no further steps were taken beyond association to prove causation.

Similarly, another study "Findings on the Central Auditory Functions of Endemic Disease Control Agents"[3] investigated occupational pesticide (and noise exposure) among adult males and found a "significant association between pesticide- and noise-exposure and the abnormal results [regarding hearing loss]." We aim to investigate this effect and further the findings of previous studies by specifically assessing a causal relationship the negative effects of at-home insecticides.

This project aims to use the most recent data from the National Health and Nutrition Examination Survey, the period of August 2021 to August 2023, to see if there is a causal relationship between frequent household insecticide use and health measures, namely hearing condition and blood pressure. Results from this study can and should be used to make recommendations to consumers on insecticide use, while also motivating scientists to conduct further research into safer alternatives for pest control.

[1] Carson, Rachel. Silent Spring. Penguin, 2002.

[2] Glover, Frank, et al. "The Association between Organophosphate Insecticides, Blood Pressure Dysregulation, and Metabolic Syndrome among U.S. Adults: NHANES 2015–2016." Hygiene and Environmental Health Advances, vol. 4, 2022, article 100035. https://doi.org/10.1016/j.heha.2022.100035

[3] de Souza Alcarás PA, Zeigelboim BS, Corazza MCA, Lüders D, Marques JM, de Lacerda ABM. Findings on the Central Auditory Functions of Endemic Disease Control Agents. Int J Environ Res Public Health. 2021 Jul 1;18(13):7051. doi: 10.3390/ijerph18137051. PMID: 34280998; PMCID: PMC8297216.

2 Data

The data used for this study is from the National Health and Nutrition Examination Survey ([4]), using their latest complete period of August 2021 to August 2023. Our treatment is `insecticides`, which is a binary outcome on whether the participant used household insecticides within the past seven days. Our first group of outcomes was hearing condition, self-reported from 6 possible options. These were one-hot encoded for testing.

The next outcome variable group is blood pressure, measused with systolic and diastolic; NHANES collects 3 measurements of each approximately 60 seconds apart. In order to truly assess the effect of insecticide use on blood pressure, we collected four outcome variables: `systolic_avg`, `diastolic_avg`, `raw_difference`, the difference between the systolic and diastolic averages, and lastly `mean_bp` calculated as: diastolic_avg $\cdot \frac{1}{3}$(systolic_avg $-$ diastolic_avg).

Only people with relevant data about insecticides, blood pressure, and hearing conditions were kept as part of the data. This was joined with covariate information, matching on individual's unique `SEQN` number. Covariate information about Demographics, Alcohol Use, Diabetes, Cholesterol, Income, Physical Activity, Sleep, Smoking, and Weight were included. These were recoded for readability, and all non-answers (NHANES codes 7, 77, 99, 999, etc.) were marked as `NA`.

At the end, there were 7,465 rows with information about insecticide use, blood pressure, and hearing condition. The corresponding chosen covariate information, based on exploratory data analysis, had many missing values demonstrated by Table 1. Many covariates were also correlated with each other, for example hours slept during the weekday vs. the weekend in Figure 1 which further helped determine which covariates would be kept for propensity score modeling. Of the 7,465 observations, 543 were treated (using insecticides) and 6922 were controls (not using insecticides). We checked the distribution of treated vs non-treated among multiple covariates, for example with race, in Figure 2. The distributions comparing the covariates were similar when stratifying by many of treatment vs. control.

When checking the distributions of blood pressure stratifying by treatment group, there was a small association between insecticides and blood pressure, specifically the systolic blood pressure, shown in Figure 3. An even stronger association was seen when comparing hearing condition between treated and control in Figure 4. The proportion of people with excellent hearing drops from 38.8% to 28.2% when going from control to treatment groups, implying an association between the two. These findings further research into whether the effect was causal.

3 Potential Outcomes

The aim of this study is to determine whether there is a causal relationship between the use of insecticides and two groups of outcomes: blood pressure and hearing condition. The different measures of blood pressure were systolic, diastolic, mean blood pressure, and the

[4]Centers for Disease Control and Prevention. "National Health and Nutrition Examination Survey (NHANES)." Centers for Disease Control and Prevention, https://www.cdc.gov/nchs/nhanes/ . Accessed 20 Apr. 2026.

raw difference of these two systolic and diastolic. NHANES specifically asked respondents whether or not they had used any chemical products in their home to control for fleas, roaches, ants, termites, or other insects within the past seven days. Therefore, the potential outcome for the treatment is the blood pressure measurement of the individual if they used one of these insecticides within the past seven days and the potential outcome for the control is the blood pressure measurement of the individual if they did not use one of these insecticides within the last seven days.

Similarly, we also check the relationship between insecticides and self-reported hearing condition. The possible measures of hearing were: Excellent, Good, A Little Trouble, Moderate Hearing Trouble, A Lot of Trouble, Deaf. The potential outcome for the treatment is the hearing condition of the individual if they had one of these insecticides within the past seven days and the potential outcome for the control is the hearing condition of the individual if they had did not use one of these insecticides within the last seven days.

To conduct causal analysis, the Stable Unit Treatment Value Assumption must be met. This assumption ensures that for individuals who used insecticide within the last seven days, that we observe the treated potential outcome, and for individuals who did not use insecticide in the last seven days, we observe the control potential outcome. Within this assumption two further core assumptions are implied, that there are no hidden or variant versions of the treatment and there is no interference of treatment values amongst units. For the first core assumption, this could be violated by how the survey question was asked in NHANES, multiple different insecticides may qualify as treatment and some insecticides for different bugs will have varying chemical formulas and levels of harm. Since the question asked about the "last 7 days" only, it does not measure how often the individual actually uses the insecticide and may exclude individuals from the treatment group if they apply insecticide to their home biweekly or monthly. Furthermore, people may be located in a geographic location that uses insecticide as pest control in public spaces, have regional specific insects they must control for, or interact with these chemicals in their jobs. As for non-inference, individuals may be located in the same house or in a similar area where an individual uses insecticide.

4 Methods

4.1 Imputation and Propensity Scores

Removing any rows with missing values would drop our dataset down to less than of the half observations from before. In order to fully take advantage of our dataset, we implemented imputation before running the logistic regression model to calculate the propensity scores. Imputation lets rows with missing covariates to still be assigned a propensity score, allowing them to be considered for matching later on. The imputation was done on the missing values from Table 1. We used a simple imputation scheme: calculating the mean of a column if the variable was continuous, or the mode of the column if it was categorical. This assumes data is completely missing at random since we did not model missing data in respect to other covariates. In addition, each variable that needed imputation (non-zero values from Table 1) was given a second column to add to the dataset. This column marked whether

each observation for that column had a true value or an imputed value. For example, if one observation's `marital_status` was imputed, they would have a `Married` value as 1 (mode), `Prev_Married` as 0, and `Never_Married` as 0. Their `marital_status_imputed` column would be 1.

The final logisitic regression model included covariates as one-hot encodings and a `_imputed` column if necessary, to find the probability of an observation being in the treated group (propensity score). An assumption of generating the propensity scores is ensuring the model is correctly specified, for example no non-linear or spline terms were omitted, and that the treatment assignment is independent. The propensity score is defined as:

$$e(x) := \text{pr}(Z = 1|X = x)$$

and becomes useful for causal inference when strong ignorability given X, which implies, $e(X)$ can be assumed meaning:

$$\{Y(1), Y(0)\} \perp Z|e(X)$$

$$0 < \text{pr}(Z|e(X) = e`) < 1 \forall e' \in e(X)$$

We believe this holds in our example due to the many covariates included, which realistically fully describe a person's probability to use insecticides. However, this may fail if there is an unmeasured confounder, which would violate strong ignorability. For example, urban status may be important for determining treatment assignment, yet not recorded in our dataset. For overlap, this could fail if lower SES individuals always use insecticide since their homes are poorly insulated, but higher SES does not, making SES completely deterministic of insecticide use. To check for issues of overlap, histograms of propensity scores in Figure 6 under treated and control individuals were plotted. The propensity score distributions look roughly similar, with 0 propensity scores for treated or control under 0.001. The ranges for treated propensity scores was [0.0188, 0.2546] and under control was [0.0144, 0.3247].

4.2 Matching Process and Checks for Imbalance

Using the generated propensity scores, a distance matrix of the absolute difference of logit propensity scores between individuals i and j were computed. A loose caliper of 1000 was added when the absolute difference exceeded 0.5 standard deviations. This distance matrix was used to match propensity scores in a 1:1 ratio, which assumes $e_{ij}(x)$ and $e_{ij'}(x)$ are the same in respect to observed and unobserved covariates. Furthermore, under strong ignorability propensity score matching is able to recover the randomization distribution from the block randomized designs:

$$\text{pr}(T(\mathbf{Y_Z}, \mathbf{Z}) \leq t|\mathcal{F}, \mathcal{Z}) = \frac{1}{|\Omega|} \sum_{\mathbf{w} \in \Omega} 1\{T(\mathbf{Y_Z}, \mathbf{Z}) \leq t\}$$

allowing for causal interpretation from observational data. A 1:1 matched structure was utilized, yielding 543 matched pairs, since it had the best balance in covariates after matching.

Standardized differences pre and post matching were computed to check for the imbalance of covariates. The improvement in imbalance was visually demonstrated through a

Love Plot in Figure 5 and checked analytically through large sample approximations of the randomization distribution using paired t-tests. Out of the 27 covariates used for propensity score generation and matching, only 1 had a p-value of less than 0.05 (whether or not the individual was previously married).

4.3 McNemar Wilcoxon, t-Test

Finally, once matching was finished we completed the analysis by comparing the matched individuals, treating each of the 543 matched sets as a pair. We calculated both a paired Wilcox Signed Rank Test and a paired t-test to evaluate whether there is a significant difference between the treatment and control groups for the blood pressure outcomes. We use all four of the blood pressure outcomes: `systolic_avg`, `diastolic_avg`, `raw_difference`, and `mean_bp` as continuous outcomes.

For the hearing conditions, because we are comparing paired proportional data, hearing outcome change from control to treated in each pair, we used the McNemar test. The hearing outcomes: `excellent`, `good`, `little_trouble`, `moderate_trouble`, and `lot_trouble` were tested. Because `deaf` has so few true observations (<10) total after matching, this was omitted as a possible response to test.

5 Results

The results from Table 3 reveal that there was no significant casual effect of insecticides on any of the blood pressure outcomes. This means with the data provided, even after creating matched sets of individuals, there is no evidence that household use of insecticides plays a role in various blood pressure metrics. However, we see different results for the hearing conditions, shown in Table 4. The `excellent` result shows that there is a significant difference between the proportion of people with excellent hearing among those who use insecticides and those who don't, with a p-value of 0.006. The high value of the McNemar statistic shows that this difference is likely not due to random chance. Because the natural ordering of these nominal outcomes, we can say that amongst hearing people insecticide causes the decline of hearing.

These findings of course rest on some key assumptions. We assume that there is not an unobserved confounder (not in NHANES) that is actually the cause of this phenomenon. SUTVA and strong conditional ignorability must hold. The potential outcomes must be independent of the treatment after adjusting for the covariates that we specifically observed and included in our propensity score model. That model as well must be correctly specified. Another key assumption is that overlap must hold; the treatment must not be deterministic. A paired Γ sensitivity analysis, maximizing and minimizing the upper tail probability, would reveal how strong the `excellent` finding is to unmeasured confounding.

One assumption is that matched individuals are indeed comparable, and do not have a confounder that truly determines the difference in `excellent`. A simple way to test this is using the methods in "Assessing Sensitivity to an Unobserved Binary Covariate in an Observational Study with Binary Outcome"[5] which explain how to assess robustness for

[5]Rosenbaum, P.R. and Rubin, D.B. (1983), Assessing Sensitivity to an Unobserved Binary Covariate in

an unobserved binary confounder. In this case, perhaps to assess the robustness to the unobserved variable urban, describing if one lives in an urban area. Another more involved method is described in "Amplification of Sensitivity Analysis in Matched Observational Studies"[6] where the amplification of a sensitivity analysis maps from each point in a low dimensional sensitivity analysis to a higher dimensional sensitivity analysis, allowing even more potential robustness from violations in assumptions.

6 Conclusion

In conclusion, utilizing the NHANES Data from the 2021-2023 period, we see a casual impact of household insecticides use on the having a good hearing condition and none on blood pressure. This result is seen after matching individuals from the survey into pairs on a variety of relevant covariates, which were partially imputed to allow many individuals to be potentially matched. However, the casual effect relies on core assumptions about confounding, strong ignorability, and SUTVA, though they can be tested for robustness through a sensitivity analysis.

6.1 Limitations

One major limitation from the data itself is that use of insecticide use was collected through a survey rather than through a biometric test, so individuals may have forgotten if they used insecticide recently. Also, that precluded any possibility of measurement for the amount of insecticide.

Furthermore, while the MCAR assumption for missing data made computation easier, it came with strong assumptions and ignored how other covariates may have influenced missing values. With more time, more robust missing data methods would have been utilized.

6.2 Lessons and Future Work

Throughout this project, the importance of collaborating with subject experts was apparent. Determining the outcome of choice came with some difficulty without background medical knowledge, and collaborating with clinicians would have strengthened our research question and purpose. Working with real life data also came with a plethora of lessons in the abundance of missingness and the need to evaluate the feasibility of some causal assumptions. The guidance from an expert would also allow accurate negative and positive controls to be added to the report, something to consider for a future work.

In future study, collaboration with clinicians should occur at the initial planning steps and more precise ways of measuring insecticide use should be investigated. While the effects of insecticides are still unclear, repeated studies remain important to ensure there are no hidden detrimental effects.

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an Observational Study with Binary Outcome. Journal of the Royal Statistical Society: Series B (Methodological), 45: 212-218. https://doi.org/10.1111/j.2517-6161.1983.tb01242

[6]Rosenbaum PR, Silber JH. Amplification of Sensitivity Analysis in Matched Observational Studies. J Am Stat Assoc. 2009 Dec 1;104(488):1398-1405. doi: 10.1198/jasa.2009.tm08470. PMID: 22888178; PMCID: PMC3416023.

7 Appendix: Figures and Tables

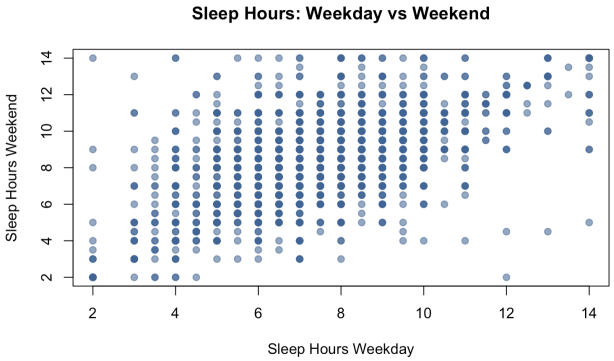


Figure 1: Sleep: Weekend vs Weekday

Variable	Missing Count
pulse_avg	0
gender	0
age_years	0
race_ethnicity_detailed	0
active_duty_military	1262
education_level	1644
marital_status	1646
household_size	0
income_poverty_ratio	942
diabetes_diagnosed	221
sedentary_minutes	1429
sleep_hours_weekday	1178
smoked_100_cigs	1395
height_inches	1172
weight_pounds	1204
mod_activity_per_week	2613

Table 1: Missing Covariate Values

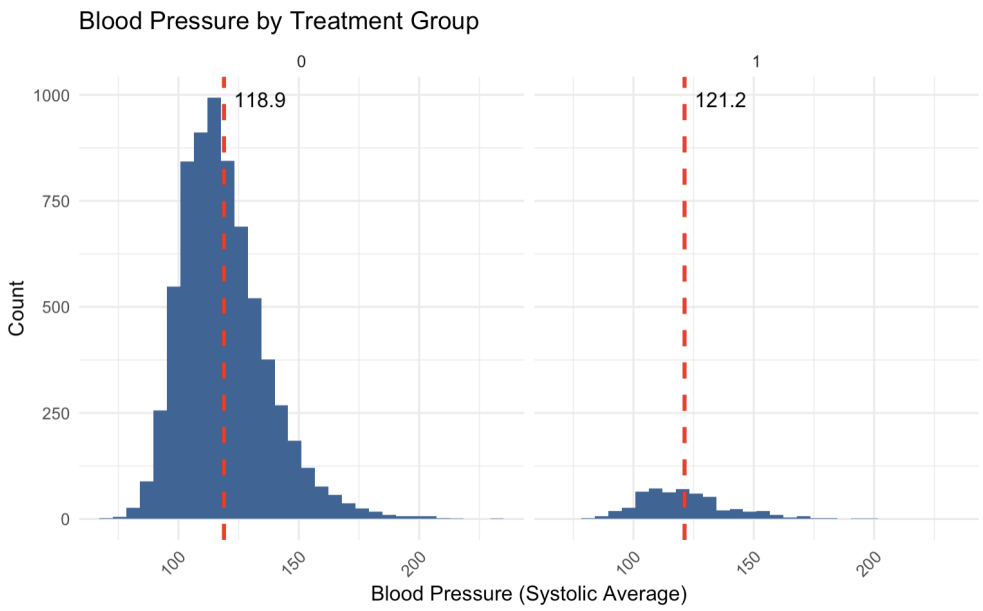


Figure 3: Blood Pressure Distribution by Treatment Group

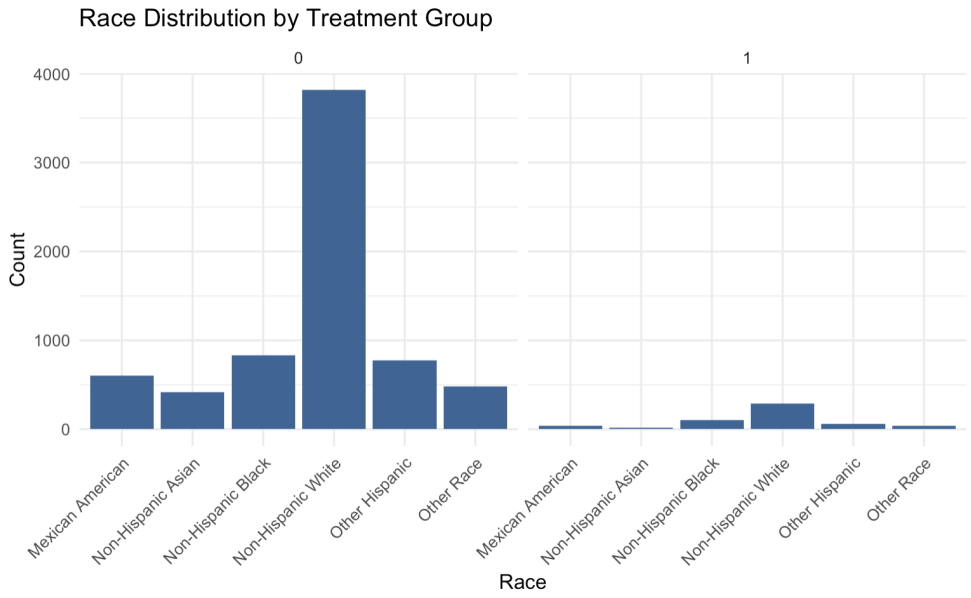


Figure 2: Race Counts by Treatment Group

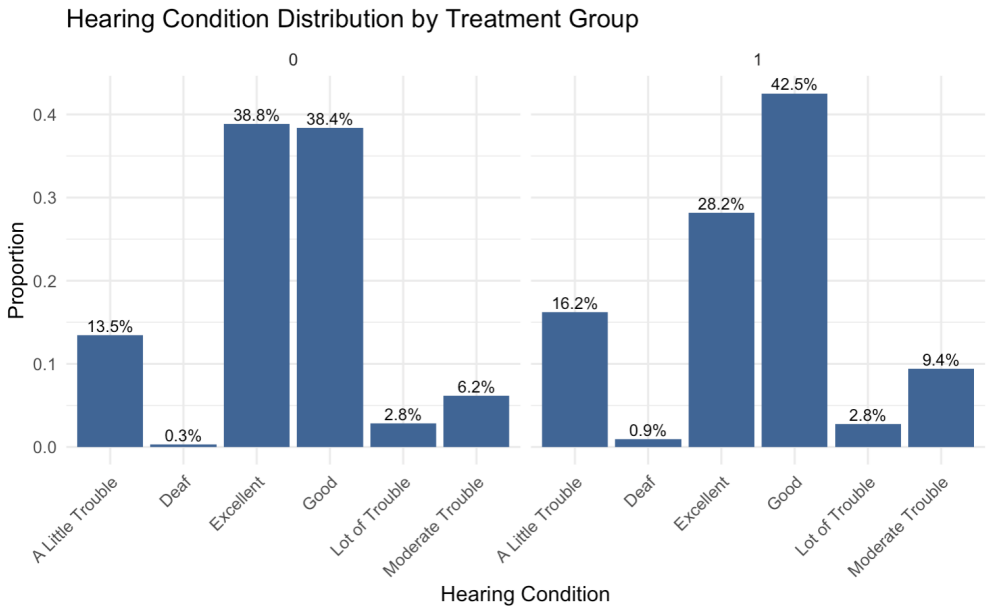


Figure 4: Hearing Condition (Proportion) Distribution by Treatment Group

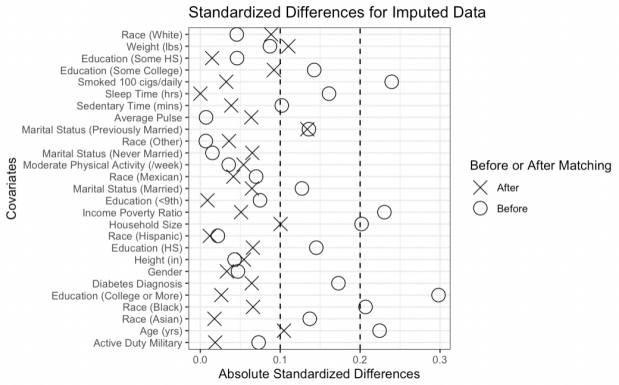


Figure 5: Love Plot with Absolute Standardized Differences.

Variable	**p-value**
Pulse Average	0.272
Gender	0.584
Age (Years)	0.071
Mexican	0.480
Asian	0.724
Black	0.334
White	0.143
Hispanic	0.837
Other	0.559
Active Duty Military	0.770
Less than 9th grade	0.891
Some High School	0.811
High School Degree	0.315
Some College	0.120
College Grad or More	0.666
Previously Married	0.029
Married	0.295
Never Married	0.291
Household Size	0.101
Income Poverty Ratio	0.380
Diabetes Diagnosed	0.333
Sedentary Minutes	0.528
Sleep Hours Weekday	0.999
Smoked 100 cigs	0.607
Height (Inches)	0.363
Weight (lbs)	0.091
Moderative Activity (per week)	0.458

Table 2: t-test after balance (paired)

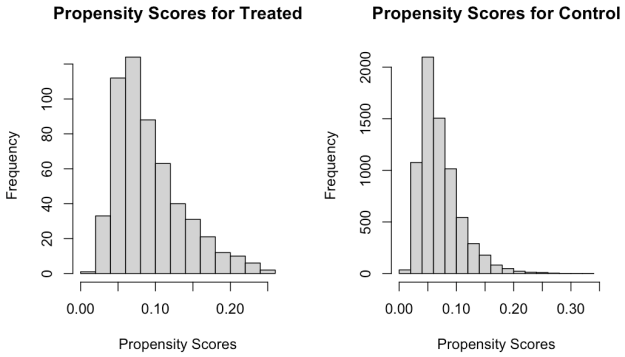


Figure 6: Propensity Scores Histograms for Checking Overlap.

Outcome	Test	95% Conf. int.	p-val
mean_bp	t-test	[-2.56, 0.37]	0.1441
mean_bp	Wilcoxon	[-2.67, 0.44]	0.1639
raw_difference	t-test	[-2.13, 1.19]	0.5781
raw_difference	Wilcoxon	[-1.83, 1.17]	0.6574
systolic_avg	t-test	[-3.58, 0.77]	0.2049
systolic_avg	Wilcoxon	[-3.50, 0.67]	0.1566
diastolic_avg	t-test	[-2.27, 0.40]	0.1692
diastolic_avg	Wilcoxon	[-1.83, 1.17]	0.6574

Table 3: Test Results for Blood Pressure Outcomes

Outcome	McNemar's chi-squared	p-val
excellent	7.5708	0.005932
good	0.64504	0.4219
little_trouble	2.5069	0.1133
moderate_trouble	3.0118	0.08266
lot_trouble	1.641	0.2002

Table 4: McNemar Test Results for Hearing Outcomes